

No Chemistry, No Causal Object

Product identity before cannabis-psychosis causality

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Scope: method paper. Not medical advice, legal advice, or treatment advice. The paper addresses exposure naming and claim size.

Abstract

The public cannabis-psychosis sentence often treats "cannabis" as if it were a stable scientific exposure. It is not. The word may point to a plant category, a legal class, a self-report item, a street label, a market, a regulated extract, resin, flower, concentrate, edible, oil, or biomarker-positive status. Those are not one dose, one route, one cannabinoid profile, or one body state.

This paper proposes the chemistry gate: no simple causal-object claim earns trust until product identity, delta-9-THC, CBD, THC:CBD ratio, route, dose, frequency, potency source, tolerance state, supply pattern, reintroduction, co-exposures, sleep, food, and service route are either measured or marked absent. Existing association findings stay in the record at their earned level. The gate blocks a different move: promotion from broad label to causal object.

Keeper:

Make the link name the drug.

1. Problem

A sentence can be grammatically small and methodologically enormous:

cannabis -> psychosis

Both nouns hide routes.

"Cannabis" can mean product chemistry, plant identity, illegal status, licensed medicine, street label, smell, possession record, use-disorder code, user identity, or survey answer. "Psychosis" can mean acute intoxication presentation, transient reaction, panic with unusual interpretation, sleep-loss state, first-episode clinical presentation, hospital admission, diagnostic code, schizophrenia-spectrum diagnosis, or family report.

A paper may study one of those pairings and write as if it studied all of them. The chemistry gate blocks that expansion.

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plant word != dose
frequency != chemistry
self report != cannabinoid profile
biomarker positive != current intoxication
diagnostic contact != incidence
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The paper does not throw away association studies. It asks each study to keep its claim at the level its exposure data can carry.

2. What the association field already contains

The association field has real feedstock. The National Academies report concluded that cannabis use is statistically associated with later schizophrenia or other psychoses, with more frequent use linked to higher association estimates. Meta-analytic work has reported a dose-response pattern by level or frequency of use. Case-control work has reported larger estimates where daily use and high-potency products are present. Experimental THC administration has produced transient psychosis-like effects in controlled settings.

Those findings matter. They do not settle the object.

A broad association variable can say:

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people in this measured category had different outcome rates
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It cannot by itself say:

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this chemical route caused this clinical object
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The missing middle is exposure identity. A causal sentence needs the drug, route, dose, timing, body state, and alternative routes.

3. The chemistry gate

The chemistry gate is a precondition for simple causal-object language.

A paper has passed the gate only when it can state, or explicitly mark missing:

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PRODUCT:
flower, resin, concentrate, edible, oil, oromucosal spray, synthetic cannabinoid,
unknown
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CHEMISTRY:
THC, CBD, THC:CBD ratio, other cannabinoids where available, contaminant screen where
relevant
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DOSE:
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per event, per day, per week, maximum event, titration pattern

ROUTE:

smoked, vaped, oral, oromucosal, mixed, unknown

FREQUENCY:

pattern across time, not only yes/no

POTENCY SOURCE:

laboratory measure, product label, police seizure average, user estimate, street name, unknown

TOLERANCE:

new user, occasional user, stable daily user, returning after break, escalating use

SUPPLY:

regulated stable product, variable street product, intermittent access, substitution suspected

REINTRODUCTION:

stop-start pattern, withdrawal period, return after sleep or appetite disruption

CO-EXPOSURES:

tobacco, alcohol, stimulants, sedatives, prescribed medicines, other drugs

BODY STATE:

sleep, food, hydration, stress load, trauma state, neurodevelopmental load

CONTACT FIELD:

policing, school or workplace action, family alarm, emergency route, diagnostic route

If those fields are absent, a study may still be valuable as a field signal. It has not earned the simple object.

4. Product identity breaks the noun

The European drug data show why the noun cannot carry chemistry. In the 2026 European Drug Report cannabis section, the European Union Drugs Agency reports that, in 2024, average THC content in EU-seized cannabis resin was 24.6 percent, about twice that of herbal cannabis at 12 percent. The same product word therefore sits over different chemical distributions across product type, country, and year.

This matters because an exposure label such as "cannabis use" can join together:

low-THC flower
high-THC flower
resin
concentrate
edible
oil
regulated extract
street product with unknown content
synthetic cannabinoid substitution
mixed tobacco product

These routes may have different onset, duration, dosing behavior, social visibility, market stability, and service-contact pattern. They should not be collapsed into one object unless the claim is explicitly about a broad category signal.

5. Frequency is not dose

Frequency is often the best available variable. It should stay. But frequency is not dose.

"Daily use" can describe:

- one low-dose evening event
- several high-THC sessions
- smoked flower with tobacco
- vaped concentrate
- edible with delayed onset
- medical oil
- unstable street supply with repeated reintroduction
- high-CBD product
- low-CBD high-THC product

Robinson et al. and Marconi et al. can support bounded statements about association by frequency or level of use. They do not convert frequency into chemistry.

The earned statement is:

- higher frequency or heavier reported use is associated with psychosis outcomes in some studies

The unearned statement is:

- a stable cannabis chemical object has been identified

The first can be tested. The second still owes product identity.

6. Route and bioavailability are not details

Route changes the object. Smoked, vaped, oral, and oromucosal routes do not have the same onset, peak, duration, titration behavior, or social context.

The Sativex Summary of Product Characteristics supplies a measurement wedge. Each 100 microlitre spray contains 2.7 mg THC and 2.5 mg CBD from Cannabis sativa L. The label states product form, route, excipients, titration, food-standardization advice, and psychiatric adverse reactions. It also distinguishes dose reduction, interruption, and monitoring routes.

That does not make Sativex the villain. It makes Sativex a contrast case.

A regulated product says:

- what it is
- how much THC
- how much CBD
- by what route
- with what dose unit
- with what titration

with what interruption rule

A loose association variable often says:

cannabis

The difference is not cosmetic. It is the difference between a drug route and a word.

7. Self-report and biomarkers answer different questions

Self-report and biomarkers are both useful. Neither is the whole exposure object.

Self-report can capture frequency, social use, perceived potency, product name, age of onset, and use pattern. It can also be affected by memory, setting, disclosure pressure, fear of sanction, current state, and the interviewer relationship. Chesney et al. emphasize that assessing cannabis use in people with psychosis requires attention to self-report limits and context.

Urine testing answers a different question. A positive THC-COOH result can show prior exposure, but it does not automatically identify current intoxication, exact dose, product, route, or time since last use. The American College of Medical Toxicology notes that positive urine results do not necessarily indicate recent use because the detection interval for THC-COOH can outlast intoxication by a wide margin.

So the repair is not:

replace self-report with urine

The repair is:

join report, product data, route data, timing data, and biomarker status without pretending any one field is the whole drug

8. Policy shapes chemistry before analysis starts

Policy is not only interpretation. Policy can shape the exposure itself.

Enforcement, legality, market route, supply stability, product testing, disclosure conditions, and service-contact pathways can all alter the measured field. A user in a regulated medical route may know product, dose, and titration. A user in an unstable street route may not know product, potency, ratio, contaminant status, or substitution.

Two UK studies are useful counterweights. Frisher et al. tested whether increasing cannabis use should have produced increasing diagnosed schizophrenia or psychoses from 1996 to 2005 and did not find the predicted population trend. Hamilton et al. found an association between cannabis reclassification and cannabis psychosis admissions in the opposite direction to the simple policy story, with policing and mental-health-system changes named among possible explanations.

Those papers do not prove a single alternative cause. They show that policy and service route can enter the measurement field.

law status -> market route -> product knowledge -> supply stability -> user disclosure -
> service contact -> diagnostic record

A chemistry gate that ignores policy can still misname the exposure.

9. Body state is part of the route

The same product can meet different bodies.

A causal object should not collapse:

first exposure
early titration
dose increase
stable dose
tolerance state
interrupted supply
return after break
withdrawal or rebound sleep/appetite/emotion disturbance
co-use weekend
sleep-debt state
stress-state use

into one line called cannabis.

D'Souza et al. show that THC can produce transient psychosis-like effects in controlled settings. That finding supports route-specific attention to THC. It does not let every cannabis word become a THC dose, nor every unusual state become a chronic psychosis object.

The route has to name the body state it is claiming.

10. Claim ladder

The chemistry gate does not make all findings vanish. It sorts claim size.

LEVEL 1: CATEGORY SIGNAL

"People marked as cannabis users differed from non-users."

Allowed when exposure is broad and marked broad.

LEVEL 2: FREQUENCY ASSOCIATION

"Heavier or more frequent reported use showed higher association estimates."

Allowed when frequency data support it.

LEVEL 3: PRODUCT-ROUTE ASSOCIATION

"Daily high-potency use, as measured or estimated in this setting, was linked to higher odds."

Allowed when product route is actually named.

LEVEL 4: CHEMICAL ROUTE CLAIM

"THC exposure by this route at this dose produced or contributed to this outcome pattern."

Allowed only with chemistry, route, dose, timing, and comparator support.

LEVEL 5: INDIVIDUAL ATTRIBUTION

"This person's episode was caused by this exposure."

Requires clinical evidence, timing, alternative-route work, co-exposure work, and caution beyond population association.

Most public sentences jump from level 1 or 2 to level 4 or 5. That is the object error.

11. Product identity instrument

A reviewer can use this instrument before accepting causal-object language.

1. What exact product was consumed?
2. Was THC measured, labeled, estimated, or unknown?
3. Was CBD measured, labeled, estimated, or unknown?
4. Was the THC:CBD ratio available?
5. Was route recorded?
6. Was dose per event recorded?
7. Was frequency recorded across time?
8. Was potency source stated?
9. Was tolerance state recorded?
10. Was stop-start or return-after-break use recorded?
11. Were tobacco, alcohol, stimulants, sedatives, medicines, and other drugs recorded?
12. Were sleep, food, hydration, and stress-state variables recorded?
13. Was product supply stable or variable?
14. Were enforcement, disclosure pressure, and service-contact route modeled?
15. Did the paper keep the conclusion at the level the exposure data earned?

Scoring rule:

- 0 to 4 fields: broad category signal only
- 5 to 8 fields: bounded association claim
- 9 to 12 fields: product-route claim possible
- 13 to 15 fields: chemical route claim may be considered

The score is not authority. It is a guardrail against noun inflation.

12. What would move the paper

The chemistry gate can be beaten by better data. The following would move a study upward:

linked product testing
event-level dose diary
route-specific recording
THC and CBD data
biomarker timing
sleep and food logging
co-exposure logging
interruption and return-after-break records
service-contact route records
pre-registered claim ladder

comparison across regulated and unregulated supply routes

The gate is therefore not anti-evidence. It is pro-rent. It tells a claim what it must pay before it can live at a larger scale.

13. Relation to the prior papers

This paper is the chemistry member of a four-paper method line.

Status-Contaminated Causality:
legal and moral labels must not behave like biological evidence

The Pair Is Not the Object:
a public noun-pair is not yet a causal object

No Chemistry, No Causal Object:
the exposure must be named before the link can be trusted

Enforcement Is an Exposure:
policy and contact fields can become part of the measured route

The shared rule is object custody:

do not let a familiar sentence become the thing it claims to measure

14. Conclusion

"Cannabis" is not one chemical quantity. It can be a plant word, a law word, a street word, a product word, a self-report word, or a biomarker word. A causal paper has to say which.

The chemistry gate keeps association findings while blocking overreach. It lets frequency studies remain frequency studies. It lets high-potency studies remain high-potency studies. It lets THC experiments remain THC experiments. It does not let the broad word "cannabis" impersonate product, dose, route, tolerance, supply, and body state.

Make the link name the drug.
Then make it name the route.
Then make it name the body state.
Then make it pay rent.

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Source custody notes

Internal draft sources used for lineage and object terms. Parent folder:
 /mnt/data/extra_extra_expanded/.

- No Chemistry exposure-state draft.
- No H0 causal-object trust draft.
- Enforcement as exposure draft.
- Confound stack draft.
- Status-contaminated causality v0.004 draft.
- SCH route-custody draft for the compressed cannabis-psychosis pair.